

Respiratory System Examination

There are four cardinal steps in examination of respiratory system.

- Inspection
- Palpation
- Percussion
- Auscultation

Inspection

This step concerns general inspection , inspection of hand and face along with precordium.

Upon general inspection we look for :

- Breathing pattern
- Use of accessory muscles (SCM)
- Flaring of ala nasi
- Audible wheeze, stridor
- Voice hoarseness
- Central cyanosis

On the hand inspection :

- Peripheral cyanosis
- Clubbing

- Asterixis (Carbondioxide retention or hypercapnia)
- Fine Tremor (If the patient is asthmatic and he was on the Beta 2 agonist such as salbutamol)

On the inspection of chest :

- Scar
- Shape of chest (Transverse : AP) or deformity in general
- Symmetry of chest during breathing
- Subcostal / Intercostal retraction
- Distended veins

Palpation

- Tenderness
- Trachea's position
- Tactile Fremitus
- Chest movement upon breathing
- Chest expansion measurement (tapemeter - might be difficult to apply in pediatrics setting)

In special cases : we might check for crepitations if we suspect subcutaneous emphysema.

Percussion

- Percussion notes
- Diaphragmatic excursion

Auscultation

- Air entry
- Breath sounds
- Adventitious sounds

And if we heard bronchial or bronchovesicular sound, we might check for transmitted voice sounds.

- Bronchophony
 - Egophony
 - Whispering pectoriloquy
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Annexures

Inspection

Breathing pattern and Respiratory rate

During inspection and vital sign measurement we have to look for rate, rhythm and pattern of breathing. And since the children are in the constant state of growth, the respiratory rate is different at different age. So based on the assigned range for a particular age group we've to interpret our findings and intervene if necessary.

Respiratory rate

Age	Respiratory rate (breath per minute)
Premature	40 - 70
0 - 3 month	35 - 55
3 - 6 month	30 - 45
6 - 12 month	25 - 40
1 - 3 years	20 - 30
3 - 6 years	20 - 25
6 - 12 years	14 - 22
12 +	12 - 18

Abnormal pattern of breathing :

- **Tachypnea** : When RR is above the upper normal range
- **Bradypnea** : When RR is below the lower normal range.
- **Apnea** : Temporary arrest of breathing
- **Kussmaul's Breathing** : Fast , deep and labored breathing which is seen in metabolic acidosis. (E.g DKA)
- **Cheyne Stokes Breathing** : Period of hyperpnea alternating with periods of apnea due to a delay in the medullary chemoreceptor response to blood gas changes (E.g. CHF and Cerebral hemispheric damage)
- **Biotic breathing** : Unpredictable, irregularity of respiration, occurs in medullary compression due to central transtentorial herniation or in conditions with elevated ICP (E.g. meningitis)
- **Apneustic breathing**: Post-inspiratory pause in breathing, occurs in pontine damage
- **Paradoxical respiration** : The abdomen sucks inwards with inspiration (it normally pouches outwards due to diaphragmatic descent) due to diaphragmatic paralysis

Signs of respiratory distress

1. Flaring of ala nasi

This happens as the patient tries to inhale and exhale as much air as possible as fast as possible owing to distress he is in.

2. Use of accessory muscles

There are muscles which are used in inspiration and expiration.

The most fundamental muscles of respiration are **Diaphragm and EIC muscle**.

Expiratory muscles are **Abdominal muscles and internal intercostal muscles**

Accessory muscles aren't used in normal respiration they come into action when TV is greater than 800 ml / breath during inspiration.

Accessory muscles are the following :

- Sternocleidomastoid Muscle
- Scalenes
- Intercostal muscles
- Serratus anterior muscles

3. Subcostal and Intercostal contraction

4. Tachypnea

Clubbing

Clubbing is bulbous enlargement of the distal segment of fingers and toes due to proliferation of connective tissue. It is suggestive of **chronic hypoxia and chronic lung disease**.

Clubbing can be graded into three.

Grade 1 : Obliteration of hyponychial angle

The normal hyponychial angle is 160. In 1st grade clubbing this angle will be greater than or equal to 180 and the normal concave deepening will be obliterated.

Grade 2 : Sponginess or softening of tissue at the nail fold

Grade 3 : Drum stick appearance of the fingers

In a 3rd grade clubbing there are different clinical parameters that could be assessed to confirm its presence beside the drum stick appearance of the nail

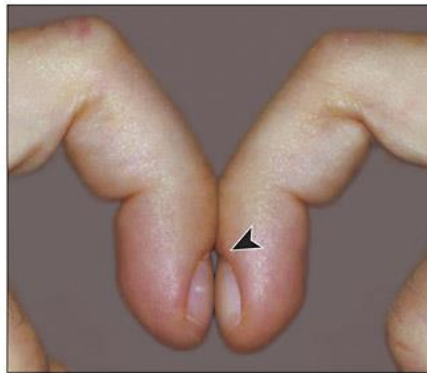
- Phalangeal depth ratio > 1

Phalangeal depth ratio = distal phalangeal depth/ interphalangeal depth

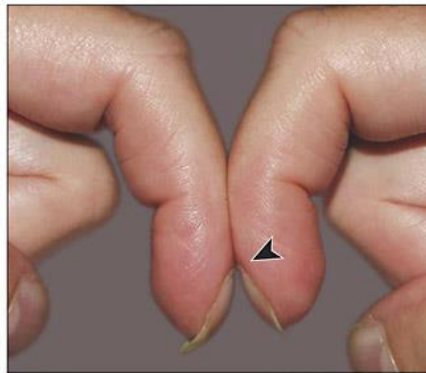
- Loss of angle of schmorh

A Schamroth sign

Normal

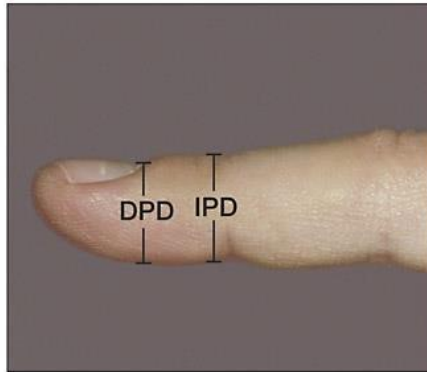


Clubbed

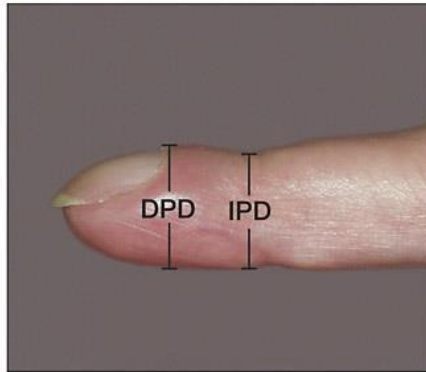


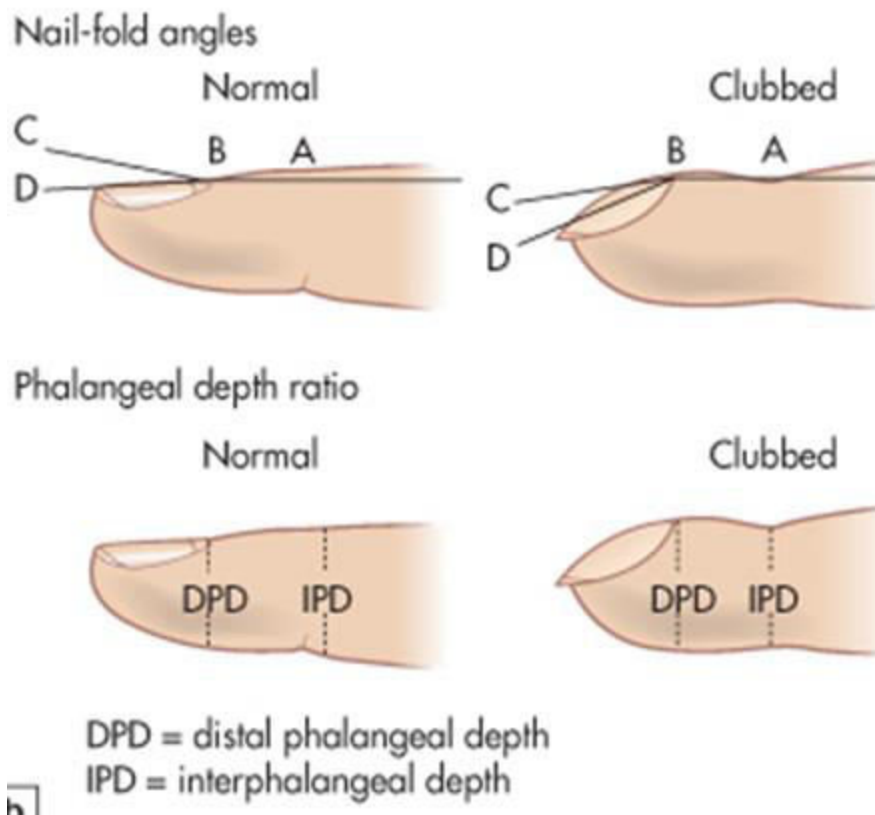
B Phalangeal depth ratio

Normal



Clubbed





Causes of clubbing are multi - systemic, but from respiratory system conditions that cause clubbing are the following :

- Tuberculosis
- Lung abcess
- Bronchiectasis
- Lung abscess
- Cystic Fibrosis
- Mesothelioma
- Primary or metastatic lung cancer

Cyanosis

Cyanosis is bluish discoloration of skin and mucus membrane due to hypoxemia resulting in increased quantity of deoxygenated hemoglobin.

Cyanosis becomes evident when the absolute concentration of deoxygenated hemoglobin is $\geq 5\text{gm/dl}$ of capillary blood or is usually obvious when arterial oxygen saturation falls below 90% in a person with a normal hemoglobin level

In patients with anemia, cyanosis doesn't occur until even greater levels of arterial desaturation is reached

Like that of clubbing, causes of cyanosis are multi - systemic, pulmonary causes include :

- COPD (Blue bloters of chronic bronchitis)
- Pulmonary embolism
- ARDS
- Advanced CF

Abnormal chest shapes

1. **Barrel chest** : Increased AP diameter of the chest in comparison to the lateral diameter seen in emphysema, chronic asthma or chronic COPD.



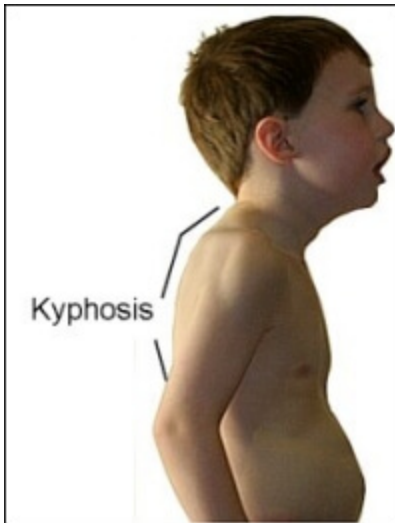
2. **Pectus excavatum** : Depression in the lower end of the sternum. This is important as it will decrease the volume of thoracic cavity.



3. **Pectus Carinatum** : aka pigeon chest - anteriorly displaced sternum with depressed costal cartilage



4. **Kyphosis** : Exaggerated forward curvature of the spine



5. **Scoliosis** Lateral curvature of the spine



6. **Harrison's Sulcus** : Linear depression of the lower ribs just above the costal margins at the site of attachment of the diaphragm often seen in *rickets* and *severe childhood asthma*



Palpation

TTT - Tenderness, Tracheal P. , Tactile Fremitus

Tenderness over the chest might suggest injury which may have resulted in rib fracture.

To assess tracheal position the 2nd and 4th fingers on each edge of sternal notch and use the 3rd finger to assess the trachea is central or deviated to one side



Slight right side tracheal deviation is normal.

If trachea is abnormally deviated to one side, it might be pushed to that side or it might be pulled to that side .

The following conditions push trachea to the opposite side :

- Mediastinal masses (Lymphoma, Thymoma and e.t.c)
- Tension pneumothorax
- Massive pleural effusion

The following conditions pull trachea to the same side of injury :

- Atelectasis
- Fibrosis
- Pneumonectomy (or extensive lobectomy)

Tactile Fremitus is assessed by putting our palm upon the anterior and posterior chest and asking the patient to say a word (specific words based on the setting - like that of a " arba arat ", "Torabati torba " or " ninety nine ") and feel for the vibration symmetrically down ward from the apices to the bases of lung.



In pediatrics we might do it when the babies cry - that would be equivalent of those specific words

Tactile Fremitus increases if there is :

- Consolidation (e.g Pneumonia)
- Parenchymal mass

Tactile Fremitus decreases if there is :

- Pneumothorax
- Pleural effusion
- Fibrosis
- Non parenchymal mass (pleural, mediastinal or chest wall)

Chest movement and expansion

During breathing the chest has to move symmetrical. If there is a lag, the affected side might have :

- Consolidation
- Pneumothorax
- Pleural effusion
- Atelectasis
- Fibrosis
- Collapse

If the chest movement is decreased symmetrically on both sides the underlying cause might be :

- Emphysema
- Bilateral lung fibrosis

Chest expansion can be measured by the total circumference of the chest along the nipple using tape measure during quiet and deep respiration (**the difference in chest expansion in deep and quiet breathing is about 4-6 cm**)

Percussion

There are three percussion notes useful to the respiratory examination

→ The normal percussion notes of healthy lung surface is **resonant**

If the percussion note becomes dull (relative), the underlying conditions are the following :

- Consolidation
- Collapse
- Mass

If the percussion note becomes stony dull, the underlying conditions are the following :

- Pleural effusion
- Hemothorax
- Empyema

If the percussion note becomes hyperresonant, the underlying condition is **Pneumothorax** .



On percussing the anterior chest, the cardiac region will be dull on the left side (or on the right if there is dextrocardia) contrasting the resonant note on other areas.

Diaphragmatic Excursion : The distance between the level of dullness on full expiration and full inspiration (normal range of diaphragmatic excursion is 5-6 cm)

Auscultation

On the auscultation, we look for breath sounds and adventitious sounds if they are present.

There are four types of breath sounds

Name of the sound	Pitch / Intensity	Duration of sound	Location where heard normally	Location where heard abnormally
Vesicular	Soft	Inspiratory longer than expiratory - there is no pause in between	Heard over most of the lung area	None - It is normal breath sound
Bronchovesicular	Intermediate	Inspiratory = Expiratory with no pause in between	→ 1st and 2nd ICS anteriorly → B/n scapulae	Anywhere else → Consolidation
Bronchial	Loud and high pitched	Expiratory > Inspiratory - with pause in between them	Over manubrium	Tubular → Consolidation → Over meniscus of pleural effusion → Collapse

				Amphoric → Cavities in the lung
Tracheal	Very loud and harsh	Inspiratory ~ = Expiratory	Over the trachea in neck	→ Consolidation → Mass



And if we heard bronchial or bronchovesicular sound and if we suspect consolidation, we might check for transmitted voice sounds.

- **Bronchophony**

Ask the patient to say "arba arat " or "ninety-nine." Normally the sounds transmitted through the chest wall are muffled and indistinct. *Louder voice sounds are called bronchophony*

- **Egophony**

We ask the patient to say "ee." Normally what is heard is a muffled long E sound. *If "ee" sounds like "A" there is egophony*

- **Whispering pectoriloquy**

We ask the patient to whisper "arba arat " or "ninety-nine". The whispered voice is normally heard faintly and indistinctly, if at all. *If louder, clearer whispered sounds are heard there is Whispering pectoriloquy*

Adventitious sounds

1. Wheeze

- Produced when air flows through narrowed / obstructed airways
- High pitched, continuous, musical sound - Hissing quality

- Heard more with expiration, can be heard on inspiration
- Associated conditions :
 - Asthma
 - COPD
 - Bronchitis

2. **Stridor**

- Produced in the setting of obstructed upper airway
- Intense inspiratory musical high pitched sound
- Associated conditions :
 - Foreign body
 - Croup
 - Epiglottitis

3. **Rhonchi**

- Produced when there is obstruction of larger airways by secretion
- Low pitched, continuous, musical sounds with snoring quality
- *Clears upon coughing*
- Associated conditions :
 - Bronchiectasis
 - Cystic Fibrosis

4. **Crackles**

- Produced when the air moves through secretion (Coarse) or air opening the collapsed alveoli (Fine).
- Discontinuous, intermittent, non musical sound
- Heard more with inspiration
- Associated conditions :
 - Pulmonary edema

- Pneumonia
- Pulmonary fibrosis
- Chronic bronchitis
- Bronchiectasis

To be commented on by & discussed with Dr. Elham Sani.

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